Non Constructible Change

AlgoExpert · Easy · Arrays

The problem

- Given positive coin values, find the minimum amount that cannot be formed.
- Sort; if the next coin exceeds (reachable + 1), reachable + 1 is the answer.

Intuition

- Sort the coins.
- If you can already build every amount up to some total, a coin bigger than total plus one leaves a gap you can never fill.

Brute force vs optimal

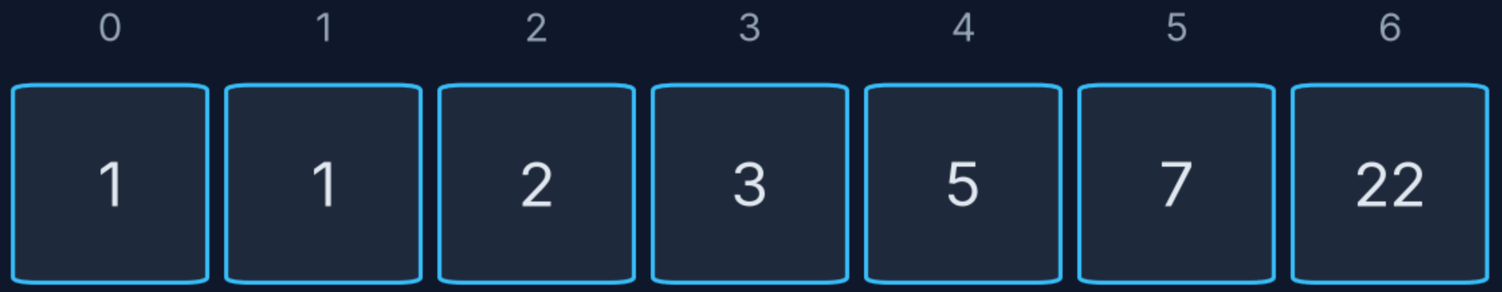
- Brute force: try amounts 1, 2, 3… and check each with subset sums
- Each check is exponential (2^n subsets)
- Optimal: sort once, then one greedy pass → O(n log n)

Approach

- 1. Sort the coins ascending
- 2. Track change, the largest amount you can build so far
- 3. If a coin exceeds change plus one, return change plus one
- 4. Otherwise add the coin to change

Worked example

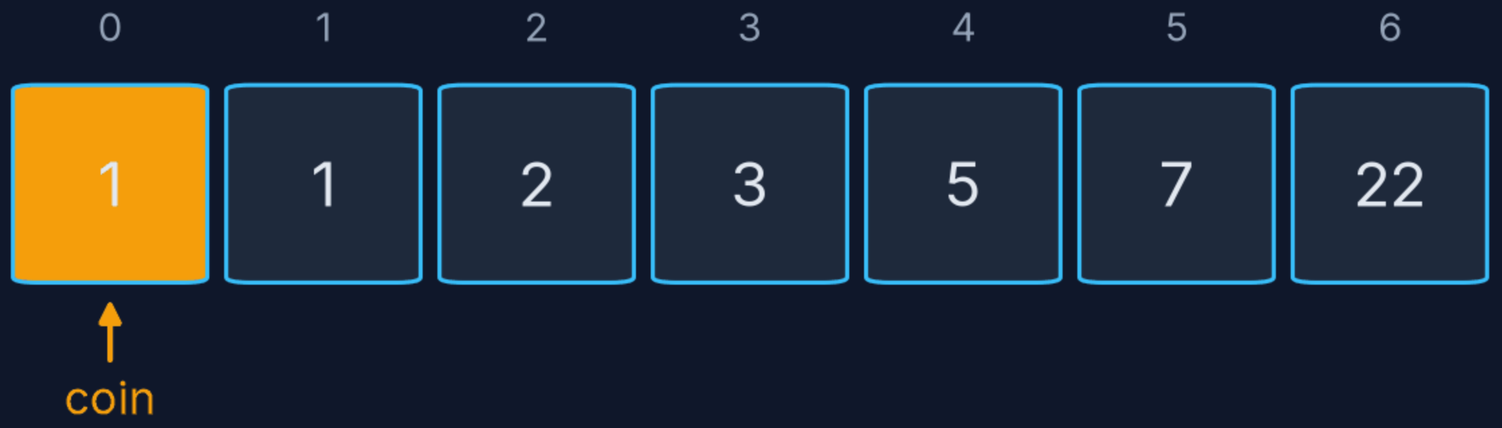
sorted coins




```
change = 0  (we can make every amount up to 0)
```

Coins: [5, 7, 1, 1, 2, 3, 22] → sorted

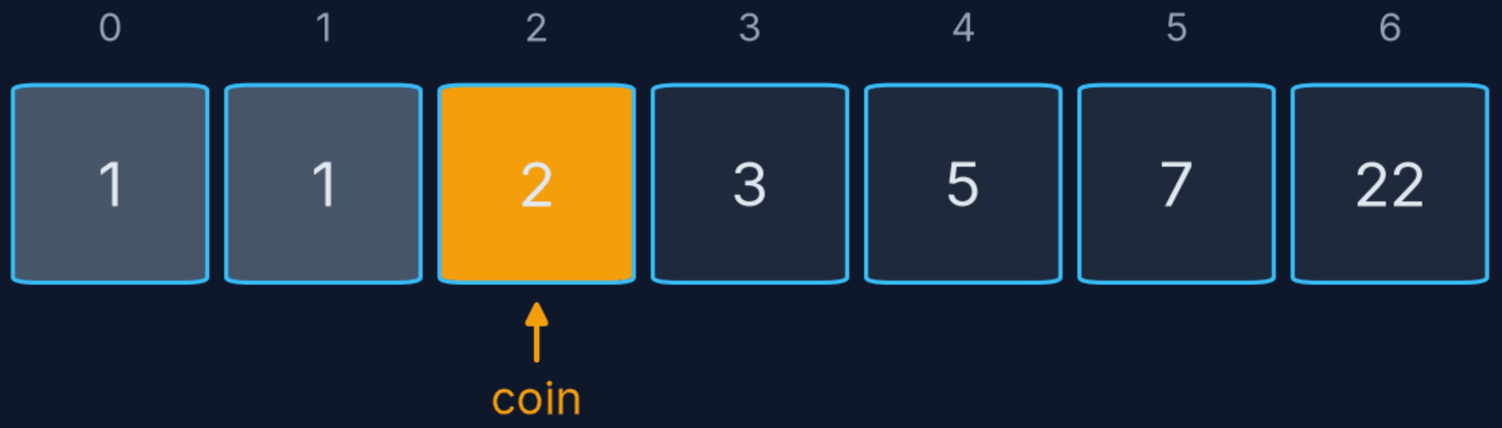
Worked example




```
change before = 0
1 ≤ 0 + 1 → change = 1
```

Every amount 1...1 is now reachable

Worked example




```
change before = 2
2 ≤ 2 + 1 → change = 4
```

Every amount 1...4 is now reachable

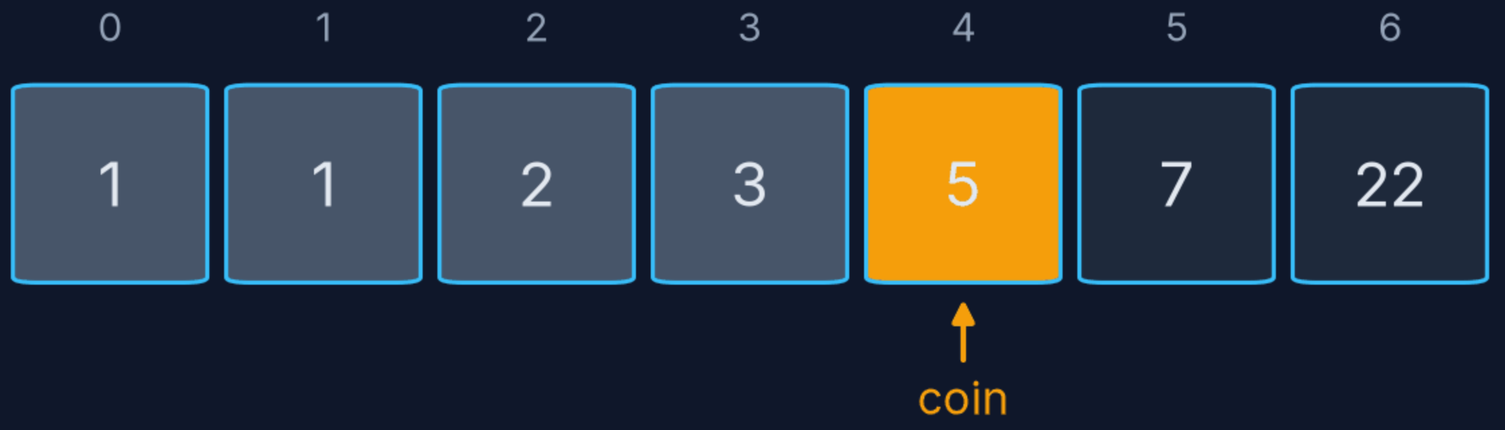
Worked example




```
change before = 4
3 ≤ 4 + 1 → change = 7
```

Every amount 1...7 is now reachable

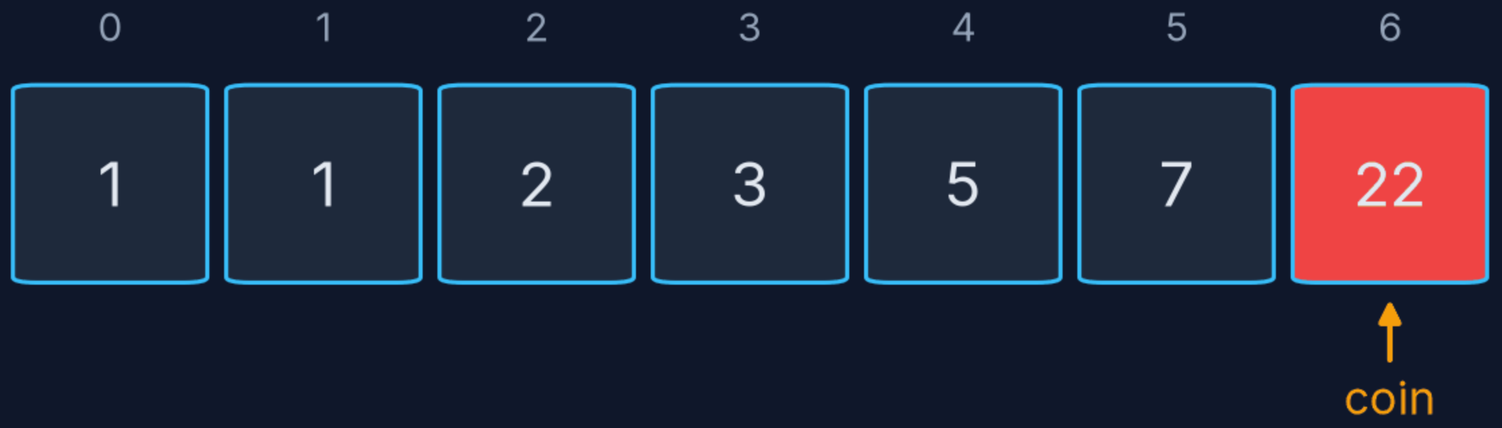
Worked example




```
change before = 7
5 ≤ 7 + 1 → change = 12
```

Every amount 1...12 is now reachable

Worked example




```
change = 19
coin 22 > 19 + 1
```

Coin 22 > 20: we can never make 20

Java solution

```java
public class NonConstructibleChange {
    public static int nonConstructibleChange(int[] coins) {
        Arrays.sort(coins);
        int reach = 0;
        for (int c : coins) {
            if (c > reach + 1) break;
            reach += c;
        }
        return reach + 1;
    }
}
```

Complexity

- Time: O(n log n)
- Space: O(1) (in-place sort)

Edge cases & pitfalls

- No coins → answer is 1
- Duplicates are fine (1, 1 → reach 2)
- Must sort first; unsorted order breaks the invariant
- Use long if values can be large

Interview tips & follow-ups

- Explain the invariant: all of 1...change are buildable
- Follow-up: what if coins can be reused? (answer changes)
- Mention it's a greedy proof by induction

Recap

- Sort the coins.
- Complexity: O(n log n)